

GENAI HANDS-ON PORTFOLIO WEBSITE CHALLENGE

Student Exercise / Question Paper - Beginner Friendly / Non-Technical

Student Name: _____ Team No.: _____
Class / Section: _____ Date: _____

Duration: 2 Hours Total Marks: 100 Final Output: One working index.html file

Instructions

- This exercise is designed for non-technical students. Prior HTML or programming knowledge is not required.
- Use Generative AI as your coding assistant. Ask it to generate, explain, improve and correct the HTML for you.
- Your final website must be a static resume-style portfolio that opens directly in a web browser.
- Use only HTML with embedded CSS in one file. Do not use APIs, databases, login systems, frameworks, package installations or server-side code.
- Do not blindly copy AI output. Review the content, test the website and ask the AI to explain anything you do not understand.
- Never paste unnecessary personal, confidential or sensitive information into a public GenAI tool.
- Use the companion Excel workbook for the raw profile dataset and final submission checklist.

Challenge Scenario: Create a Personal Portfolio Website

You have been asked to create a professional online portfolio for Aarav Mehta, a college student preparing for internships and future job opportunities. The portfolio should work like a simple digital resume and should present education, skills, projects, experience and achievements in an attractive, easy-to-read format.

You are not expected to write the website manually from scratch. Your challenge is to use GenAI effectively, safely and responsibly to produce the website.

Task 1 - Plan the Portfolio [10 Marks]

- Identify the main purpose of the portfolio and its intended audience.
- Decide the website sections that should be included.
- Define at least three design goals, for example: professional, simple, readable, modern, mobile-friendly.
- Decide what information should NOT appear on a public portfolio.

ACCEPTANCE CRITERIA

- ✓ Purpose and audience are clearly stated.
- ✓ At least 5 website sections are planned.
- ✓ At least 3 design goals are documented.
- ✓ At least 2 examples of information that should not be public are identified.

Task 2 - Ethical & Responsible AI: Clean the Personal Data [20 Marks]

Dataset: Open the companion Excel workbook and use the sheet Portfolio Profile - Raw. It contains 20 profile fields, including deliberately included personal and sensitive information.

Important: Do NOT copy the entire raw dataset into a public GenAI tool.

- Review all profile fields and identify information that is safe for a public portfolio.
- Remove, mask, generalize or replace unnecessary sensitive information before using the profile with GenAI.
- Pay special attention to phone number, personal email, exact home address, date of birth, government ID and other identifiers.
- Create a cleaned, privacy-safe profile that still contains enough information to build a professional portfolio.
- Write three privacy rules you would follow when creating a public portfolio using GenAI.

ACCEPTANCE CRITERIA

- ✓ All 20 fields are reviewed.
- ✓ Sensitive/unnecessary information is removed, masked, generalized or replaced before GenAI use.
- ✓ No government ID, exact home address or other high-risk personal data is included in the AI prompt or final website.
- ✓ The cleaned profile still contains sufficient professional information.
- ✓ 3 privacy rules are documented.

Task 3 - Prompt Engineering: Ask GenAI to Build the Website [15 Marks]

First try this simple prompt:

Basic Prompt: "Create a portfolio website for this person using HTML."

Then create a stronger prompt. It should tell the AI:

- Its role - for example, beginner-friendly web developer.
- The purpose and audience of the portfolio.
- The cleaned profile information from Task 2.
- The required website sections.
- Design expectations and constraints.
- That the result must be one complete index.html file with embedded CSS.
- That the code should be simple and include helpful comments so a beginner can understand it.

ACCEPTANCE CRITERIA

- ✓ Improved prompt includes role, context, cleaned data, required sections, design requirements, technical constraints and expected output.
- ✓ Only privacy-safe profile data is included.
- ✓ Basic and improved prompts are both retained for comparison.
- ✓ At least 3 improvements in the second AI response are identified.

Task 4 - Generate the First Working Website [15 Marks]

Use your improved prompt to generate the first version of the website. Save the AI-generated code as index.html and open it in a browser.

- Ask GenAI for one complete HTML document, not separate code fragments.
- If you do not know how to save HTML, ask GenAI to explain the steps in simple language.
- Open the file in Chrome, Edge or another browser and check whether it displays correctly.

ACCEPTANCE CRITERIA

- ✓ Exactly one main file named index.html is produced.
- ✓ The file opens locally by double-clicking it; no server, login, API key or installation is required.
- ✓ HTML and CSS are contained in the same file.
- ✓ The page loads without displaying raw code or obvious errors.

Task 5 - Portfolio Content Requirements [10 Marks]

Review the generated portfolio and make sure the content is useful, accurate and suitable for a public professional profile.

- Include a clear name/title or hero section.
- Include About / Profile Summary.
- Include Education and Skills.
- Include Project and/or Internship/Experience.
- Include Achievement and/or Interests.
- Include a safe Contact section or professional social link only where appropriate.

ACCEPTANCE CRITERIA

- ✓ Portfolio contains the required professional sections.

- ✓ Content matches the cleaned profile data and does not invent qualifications or experience.
- ✓ No sensitive information removed in Task 2 reappears in the website.
- ✓ Spelling, grammar and professional tone are reviewed by the student.

Task 6 - Improve the Design Using GenAI [10 Marks]

Ask GenAI to improve the website without making it technically complex.

- Improve spacing, headings and visual hierarchy.
- Use consistent fonts and a professional color scheme.
- Add simple navigation links to the major sections.
- Make the layout readable on both desktop and mobile screens.
- Optional: ask for simple hover effects or visual cards, but keep the website static and beginner-friendly.

ACCEPTANCE CRITERIA

- ✓ Website has clear spacing and readable sections.
- ✓ Navigation is easy to understand.
- ✓ Design is visually consistent.
- ✓ Website remains usable on a smaller/mobile browser window.
- ✓ No unnecessary complex library or framework is introduced.

Task 7 - Test, Find Problems and Fix Them [10 Marks]

Do not assume AI-generated code is correct. Test the website yourself and use GenAI to help fix problems.

- Check that the page opens correctly.
- Check navigation links.
- Check all text against the cleaned profile.
- Check readability and spacing.
- Resize the browser and check the mobile layout.
- If you find a problem, describe it to GenAI and ask for a corrected complete HTML file.

ACCEPTANCE CRITERIA

- ✓ At least 5 test checks are documented.
- ✓ Any identified problem is corrected or clearly explained.
- ✓ Final HTML is tested again after changes.
- ✓ Student can explain at least one change made between the first and final version.

Task 8 - Hallucination, Copyright, Privacy & Human Verification [5 Marks]

Review what the AI created and answer:

- Did the AI add any skill, achievement, job experience, link or qualification that was not provided?
- Did it reintroduce any sensitive information?
- Did it use external images, content or links that you cannot verify?
- What information must a human verify before publishing the portfolio?
- Identify at least three AI-generated items or decisions that required human review.

ACCEPTANCE CRITERIA

- ✓ Hallucination risk is considered.
- ✓ Privacy is rechecked before publication.
- ✓ External/unverified content is identified where applicable.
- ✓ At least 3 items requiring human verification are documented.

Task 9 - Final Submission & Demonstration [5 Marks]

Submit the final index.html and prepare a short 3-5 minute demonstration.

- Open and demonstrate the final portfolio website.
- Show the improved prompt used to generate it.
- Explain one privacy decision you made.
- Explain one design/code improvement made using GenAI.
- Explain one AI suggestion or output you rejected/corrected and why.

ACCEPTANCE CRITERIA

- ✓ Final index.html opens and works locally.
- ✓ Website is suitable for a public resume-style portfolio.
- ✓ Demonstration covers prompt engineering, privacy, improvement and human judgement.
- ✓ Submission Checklist in the companion Excel is completed.

Final Submission

Complete the Submission Checklist sheet in the companion Excel workbook before submitting your work.

Final Reminder: You do not need to become a programmer for this exercise. The objective is to learn how to instruct GenAI clearly, protect personal information, evaluate AI-generated output, test the result and use human judgement before publishing.